Syllabus for Fluid Mechanics I

DESCRIPTION

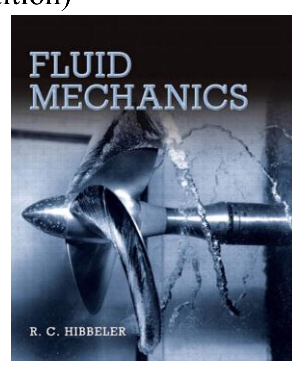
Fluid mechanics is a fundamental engineering course that describes the properties of fluids, principles of fluids in static conditions and motion, and the physical processes involved in fluid flow through conduits. In particular, the fluid in focus for this course is water for civil engineering applications. This course will introduce the students to Newtonian and Non-Newtonian fluids, the forces on a fluid in a stationary state (statics) with applications related to buoyancy, as well as fluids in motion (kinematics). Flow in conduits with application in pipe flow will introduce the students to Bernoulli's principle for steady-state applications. The energy equation will be discussed with a discussion about major and minor losses in pipes. Laboratory experiments will be used to demonstrate the concepts discussed in the classroom. Numerical computational models and software will also be used to apply the principles learned in class. Sustainable engineering practices and global issues will be discussed.

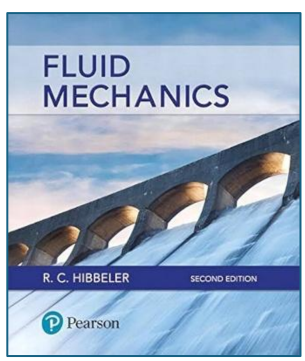
INSTRUCTORS

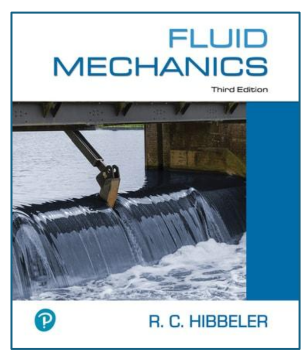
Yingxiao Liu, Ph.D (liuyin@rowan.edu)
Engineering Hall, Room 227

TEXTBOOK

Fluid Mechanics
Authors: Russell C. Hibbeler
Publisher: Pearson
ISBN 978-0132777629 (First Edition); 978-0134649290 (Second Edition); 978-0137839490 (Third Edition)







Textbook is recommended and not required.

LEARNING OUTCOMES

1. Define a fluid and describe its fundamental properties
2. Describe the concepts of fluid statics and kinematics
3. Describe the hydraulic concepts involving pressurized pipe flow
4. Apply the concepts of pressurized pipe flow for pipe sizing and design
5. Demonstrate the use of computational tools for the analysis of pressurized and unpressurized steady water flow

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FINAL GRADING BREAKDOWN

- Attendance 10%
- Midterm 20%
- Laboratory 20%
- Homework 20%
- Final Exam 30%

Extra credit opportunities will be offered during the semester to enhance student learning. These opportunities will be open to all students and will account for no more than 10% of the total course grade. Examples include optional homework problems, short reflection essays, or seminar participation. Extra credit is intended to supplement learning and cannot replace consistent performance on homework, labs, and exams.

POLICIES

- **Lecture Attendance** is mandatory. If you know that you will be absent from a class for a valid, documented reason (medical, family emergency, or other reasonable circumstances), obtain approval from the instructor 24 hours before the class period.
- **Lab Attendance** is mandatory. Missing any lab session without a valid reason will result in a zero grade for that lab session. If the absence is due to a valid reason, the student must arrange to complete a makeup lab and submit their own lab report. Students who miss more than two lab sessions without valid reasons will receive a zero grade for the entire lab portion (20% of the final grade).
- **Homework** must be submitted by the posted due date and time. Late homework will be accepted only under valid, documented reasons. Without a valid excuse, the grade of each homework will be reduced by 33% for each day past the due date. Homework submitted more than 3 days late without a valid excuse will receive a grade of zero.
- **Exams** must be taken at the scheduled times. No makeup exams will be given except in cases of a valid, documented reason. Missing any exam without a valid, documented excuse will result in a zero grade for that exam.
- **Classroom Respect and Inclusivity**. This classroom is a place where you will be treated with respect. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual

orientations, abilities, and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming, and inclusive environment for everyone. If at any time you feel that your contributions are not being valued, please speak with me privately. If you prefer to communicate anonymously, you may do so in writing or by contacting the Office of Social Justice, Inclusion, and Conflict Resolution (https://rowan.campuslabs.com/engage/organization/sjicr).

- **Accommodations**. Your academic success is important to me. If you have a documented disability that may affect your work in this class, please contact me as soon as possible so we can discuss your needs. To receive official University services and accommodations, students should provide documentation of their disability to the Office of Accessibility Services. The staff there are available to answer questions about accommodations and to assist you in the process.
- Students are expected to comply with the **Academic Integrity Policies** of Rowan University. Any violations of academic integrity will be reported to the University's Provost Office for further action in accordance with University procedures.

TENTATIVE LECTURE SCHEDULE

Engineering Hall 320
Tuesday 11:00 am - 12:15 pm

Dates	Topics
Sep 2	Introduction
Sep 9	Fluid Properties
Sep 16	Fluid Properties
Sep 23	Fluid Statics
Sep 30	Fluid Statics
Oct 7	Fluid Statics
Oct 14	Kinematics and Bernoulli
Oct 21	Kinematics and Bernoulli
Oct 28	**Midterm Exam**
Nov 4	Kinematics and Bernoulli
Nov 11	Pipe Flow
Nov 18	Pipe Flow
Nov 25	Pipe Flow
Dec 2	Pumps & Turbines
Dec 9	Recap
Dec 11 – Dec 17	**Final Exam**

LAB INFORMATION

Professional conduct is required in the laboratory to avoid safety violations. Online safety training is mandatory for all students. Some of the important things to remember:

- No food or drink is allowed in the laboratory

- Pants and closed-toe shoes are required to enter the lab
- **All lab reports are team reports unless specified**
- Lab reports are due before each lab class
- Lab grades will be weighted based on the contribution percentage given by your teammates

LAB REPORT All lab reports should include the following sections.

1. **Title**
2. **Abstract** – A brief paragraph summarizing the experimental procedures and results
3. **Introduction** – A general background should be presented here with relevant citations for statements. The objective of the report needs to be at the end of this section, and it is usually one sentence describing the goal of the experiment.
4. **Materials and methods** – This section will include a list of all the materials and instruments that were used in the experiment. The method of the experiment should be presented in detail. If it is a software lab, the input parameters should be discussed in this section. Any formula used for generating results should be presented here. All the parameters in the formula should be explained in detail. All the equations or formulae should be numbered.
5. **Results and discussion** – This section should include the results from the experiment. The results should be discussed in detail and should refer to the relevant formula for calculations. Figures and tables should be accompanied by results where necessary. The figure captions go under the figure, and the table captions should go above the Table. Uncertainty and appropriate significant figures should be used where necessary.
6. **Conclusions** – This section should briefly summarize the findings from the experiment.
7. **References** – This section will contain the material used to write the lab report. There is no preferred format for citation as long as it stays consistent throughout the report.

TENTATIVE LAB SCHEDULE

Engineering Hall 118

Section 1: Tuesday 5:00 pm – 7:45 pm

Section 2: Wednesday 5:00 pm – 7:45 pm

Dates	Topics
Sep 2, 3	-
Sep 9, 10	Lab 1: Excel for Data Analysis
Sep 16, 17	Lab 2: Measurement of Fluid Properties
Sep 23, 24	Lab 3: Hydrostatic Force
Sep 30, Oct 1	-
Oct 7, 8	Lab 4: Impact of Jet
Oct 14, 15	-
Oct 21, 22	Lab 5: Orifice and Free Jet
Oct 28, 29	-
Nov 4, 5	Lab 6: Bernoulli's Principle
Nov 11, 12	Lab 7: Flowmeter Calibration
Nov 18, 19	Lab 8: Energy Losses in Pipes

Nov 25, 26	-
Dec 2, 3	-
Dec 9, 10	-
Dec 16, 17	-